

Dietary intake of meat and meat-derived heterocyclic aromatic amines and their correlation with DNA adducts in female breast tissue.

It was the aim of this study to examine the association of the consumption of meat in general, meat prepared by different cooking methods and the dietary intake of heterocyclic aromatic amines (HCA) with the level of DNA adducts in the breast tissue of women undergoing reduction mammoplasty. Dietary intake of meat and HCA were assessed via questionnaire in 44 women undergoing reduction mammoplasty. DNA adduct analysis in breast tissue was performed by (32)P-postlabelling analysis. Spearman rank correlation coefficients (r) were calculated to examine the association of meat consumption and dietary HCA intake with tissue DNA adduct levels. A median DNA adduct level of 18.45 (interquartile range 12.81-25.65) per 10(9) nucleotides in breast tissue was observed; median HCA intake was 40.43 ng/day (interquartile range 19.55-102.33 ng/day). Total HCA intake ($r = 0.33$, $P = 0.03$), consumption of fried meat ($r = 0.39$, $P = 0.01$), beef ($r = 0.32$, $P = 0.03$) and processed meat ($r = 0.51$, $P = 0.0004$) were statistically significantly correlated with the level of DNA adducts in breast tissue. The detected DNA adducts could not be confirmed to be specific HCA-derived DNA adducts by comparison with external standards, using the (32)P-postlabelling assay. We observed strong correlations of dietary HCA intake and consumption of fried and processed meat with DNA adduct levels in breast tissue of 44 women. Since the detected DNA adducts were not necessarily specific only for HCA, it is possible that HCA intake is a surrogate of other genotoxic substances, such as polycyclic aromatic hydrocarbons, in meat prepared at high temperatures.